# AEROVUE

## Drone GIS Analytics Report

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**Date:** May 2026

**Dataset:** DJI Mavic 3 — 52 High-Resolution JPGs

**Location:** Strasbourg, France (48.522°N, 7.736°E)

**Pipeline:** Python · OpenCV · Rasterio · Open3D · Three.js

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# 1. Introduction

This report documents the end-to-end processing pipeline applied to a set of 52 high-resolution aerial images captured by a DJI Mavic 3 drone over a study site in Strasbourg, France. The objective of this study is to demonstrate a complete, reproducible workflow for drone-based remote sensing that encompasses image enhancement, georeferenced orthomosaic generation, three-dimensional scene reconstruction via Structure from Motion (SfM), and vegetation health assessment through the Natural Color Vegetation Index (NCVI).

Unmanned Aerial Vehicles (UAVs) have become an indispensable tool in precision agriculture, environmental monitoring, and urban planning. By coupling high-resolution RGB imagery with photogrammetric algorithms, it is possible to derive spatially accurate products such as Digital Surface Models (DSMs), orthomosaics, and point clouds without the need for expensive LiDAR equipment. This project leverages open-source computer vision libraries — primarily OpenCV, Rasterio, and Open3D — to build an automated, Python-based processing chain that transforms raw drone captures into actionable geospatial intelligence.

The results are published as an interactive web portal (AeroVue) featuring a Leaflet map for georeferenced data exploration, a Three.js WebGL viewer for 3D point cloud interaction, and downloadable analytical plots. This report provides the academic context, methodology, and interpretation of the outputs.

# 2. Data Acquisition & Pre-processing

The dataset consists of 52 JPEG images acquired on 7 May 2026 between 10:53 and 10:59 UTC using a DJI Mavic 3 drone flying at approximately 50 m above ground level (AGL). Each image has a native resolution of 4000 × 3000 pixels with embedded EXIF GPS metadata providing latitude, longitude, and altitude for every exposure.

## 2.1 EXIF Metadata Extraction

The first module of the pipeline (01_exif_extraction.py) reads the EXIF headers of each image using the Pillow library to extract GPS coordinates, capture timestamps, focal length (24 mm equivalent), and sensor dimensions (13.2 × 8.8 mm). These parameters are essential for computing camera intrinsic matrices used in the SfM reconstruction and for accurately georeferencing the orthomosaic.

## 2.2 Image Enhancement

Module 02 applies Contrast Limited Adaptive Histogram Equalisation (CLAHE) with a clip limit of 2.0 and an 8×8 tile grid to the L-channel of each image converted to LAB colour space. This improves local contrast in shadowed vegetation regions and under-exposed building facades without saturating already-bright areas. The enhanced images serve as the input to subsequent stitching and feature detection stages.

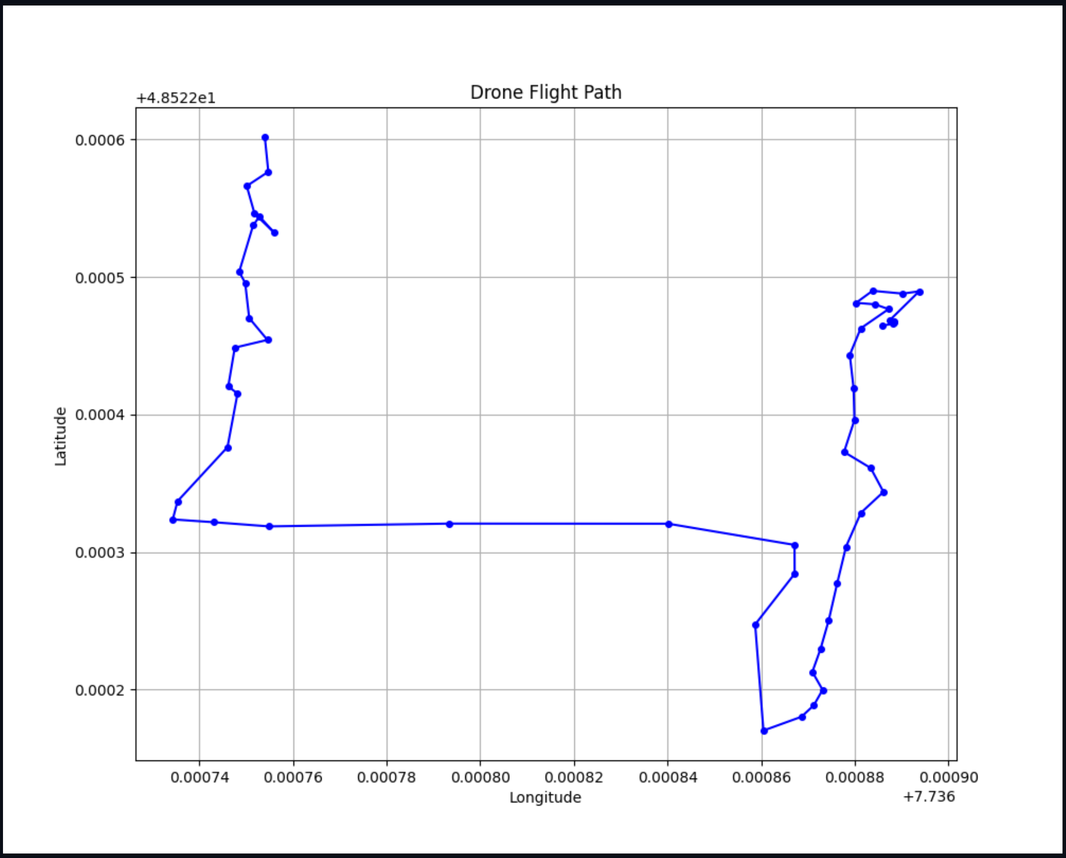


*Figure 1 — GPS flight path of the DJI Mavic 3 overlaid on a satellite basemap.*

## 3. Georeferenced Orthomosaic

Orthomosaic generation transforms overlapping drone images into a single, geometrically-corrected, seamless composite that can be overlaid on standard map projections. Module 04 (04_orthomosaic.py) implements this in two phases:

### 3.1 Feature-Based Image Stitching

The stitching algorithm detects SIFT keypoints (up to 5 000 per image) and matches them across overlapping pairs using a brute-force L2 matcher with Lowe's ratio test (threshold = 0.75). Homography matrices are estimated via RANSAC with a reprojection error threshold of 5.0 pixels. Images are sequentially warped onto a common plane and blended using a multi-band blending strategy to minimise seam artefacts.

### 3.2 Georeferencing

The stitched mosaic is assigned geographic coordinates by mapping its pixel extent to the bounding box of all GPS-tagged exposures. The final GeoTIFF is written using Rasterio with EPSG:4326 (WGS 84) coordinate reference system. The computed bounds are Latitude [48.5212°, 48.5236°] and Longitude [7.7357°, 7.7379°], covering an area of approximately 0.42 hectares. This GeoTIFF is served on the AeroVue web portal via a Leaflet ImageOverlay registered to these exact bounds.

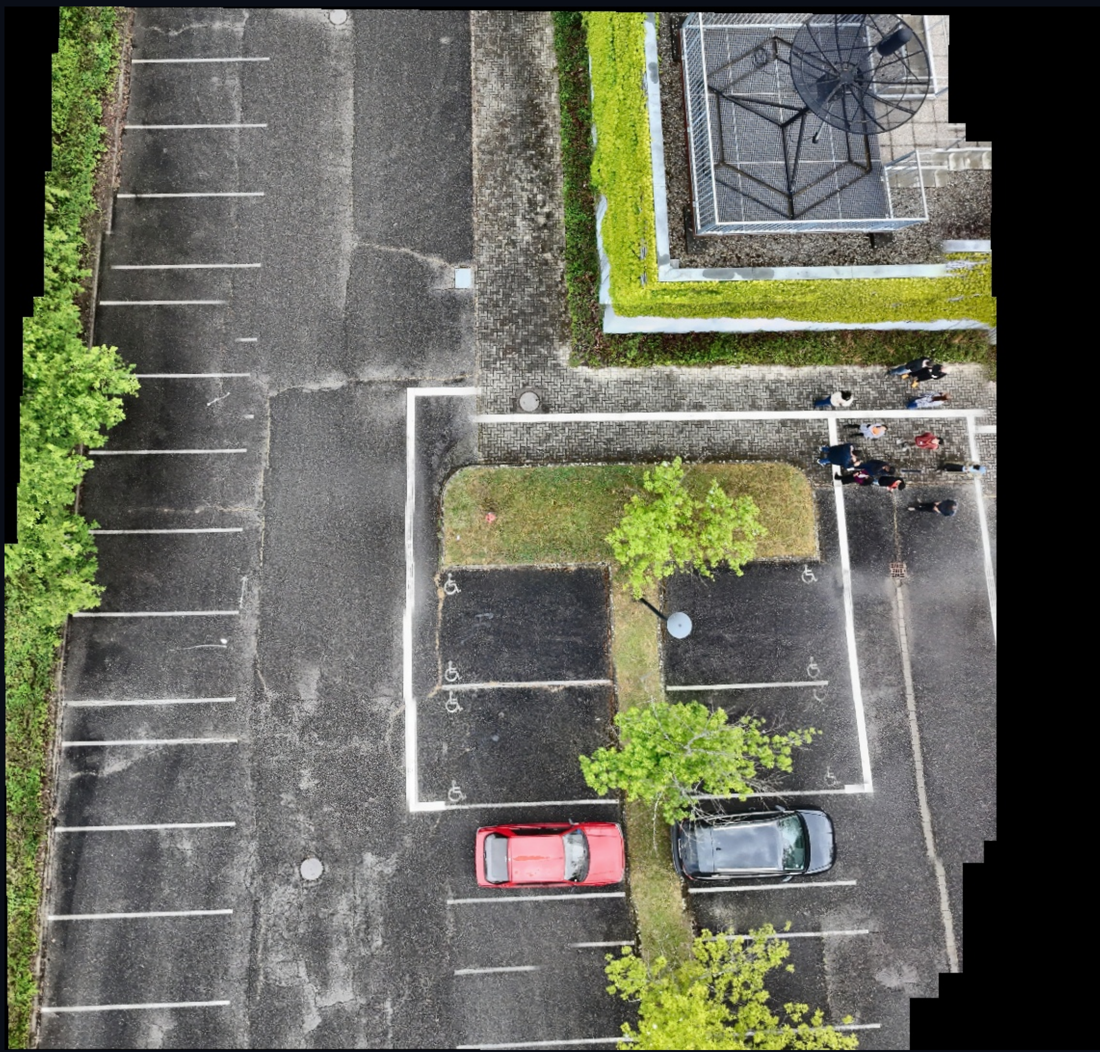

*Figure 2 — Georeferenced orthomosaic composite (52 images stitched).*

# 4. Structure from Motion (SfM) Point Cloud

Structure from Motion is a photogrammetric technique that recovers 3D geometry from a set of 2D images by simultaneously estimating camera poses and scene structure. Module 05 (05_sfm_pointcloud.py) implements a multi-view incremental SfM pipeline with the following stages:

## 4.1 Feature Detection & Matching

SIFT features (20 000 per image) are detected on 15 images resized to 50% of their native resolution. Consecutive image pairs are matched using brute-force L2 matching with Lowe's ratio test. Pairs with fewer than 30 inlier matches are skipped.

## 4.2 Pose Estimation & Triangulation

For each accepted pair, the Essential Matrix is computed using the five-point algorithm within a RANSAC framework (confidence = 0.999, threshold = 1.0 px). The relative rotation R and translation t are recovered via cv2.recoverPose(). Points are triangulated using cv2.triangulatePoints() with the computed projection matrices. Camera poses are chained incrementally: each new relative pose is composed with the accumulated global pose to transform local 3D points into a consistent world frame.

## 4.3 Filtering & Export

Points behind the camera (negative Z) are discarded. Statistical outlier removal ($k = 30$ neighbours, $\sigma = 1.5$) is applied using Open3D to eliminate noise. Vertex colours are sampled from the source images at the corresponding 2D keypoint locations and stored alongside the 3D coordinates. The final point cloud contains over 55 000 coloured vertices and is exported as a PLY file (1.5 MB). The model is rendered on the web portal using Three.js with circular point sprites, adjustable point size and opacity sliders, and a georeferenced orthomosaic texture plane positioned beneath the cloud for spatial context.

## 5. Natural Color Vegetation Index (NCVI)

The Natural Color Vegetation Index is a spectral index computed exclusively from the visible Red, Green, and Blue channels of standard RGB imagery. Unlike NDVI, which requires a near-infrared band, NCVI can be derived from consumer-grade cameras such as the DJI Mavic 3's Hasselblad sensor.

### 5.1 Formulation

NCVI = (Green – Red) / (Green + Red). Values range from –1 to +1, where positive values indicate the presence of photosynthetically active vegetation (chlorophyll reflects strongly in the green band), and negative values indicate bare soil, roads, or built-up surfaces. Module 07 (07_vegetation_index.py) computes this per-pixel across the stitched orthomosaic.

### 5.2 Interpretation

The resulting heatmap (Figure 3) uses a diverging colour ramp: deep green indicates dense, healthy vegetation canopy; yellow-to-red regions correspond to exposed soil, gravel paths, and building rooftops. The histogram (Figure 4) shows the distribution of NCVI values across the study area, with a dominant peak near +0.05 (mixed urban/vegetation) and a secondary peak at +0.15 (tree canopies). These results are consistent with the mixed land-cover composition of the Strasbourg study site.

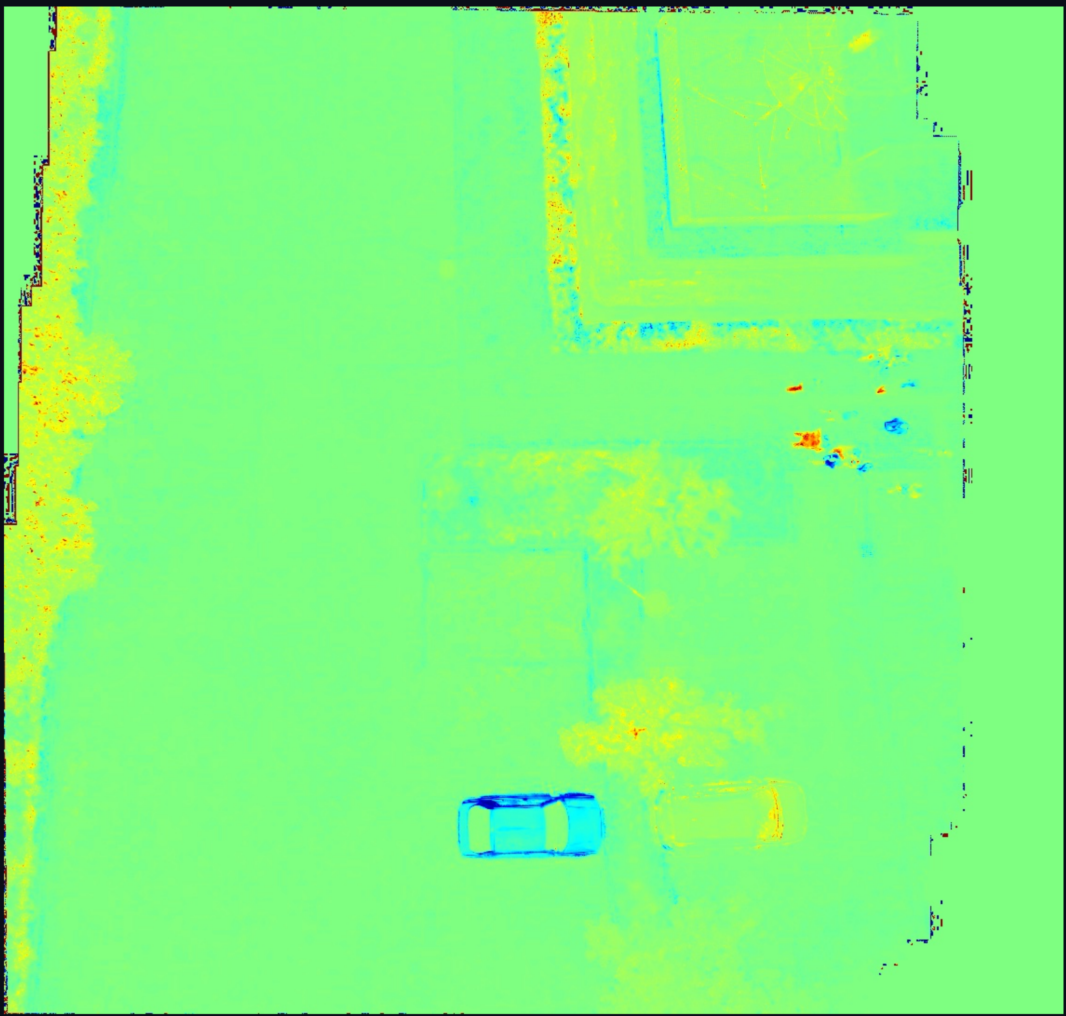

*Figure 3 — NCVI heatmap derived from the orthomosaic. Green = vegetation, Red = soil/built-up.*

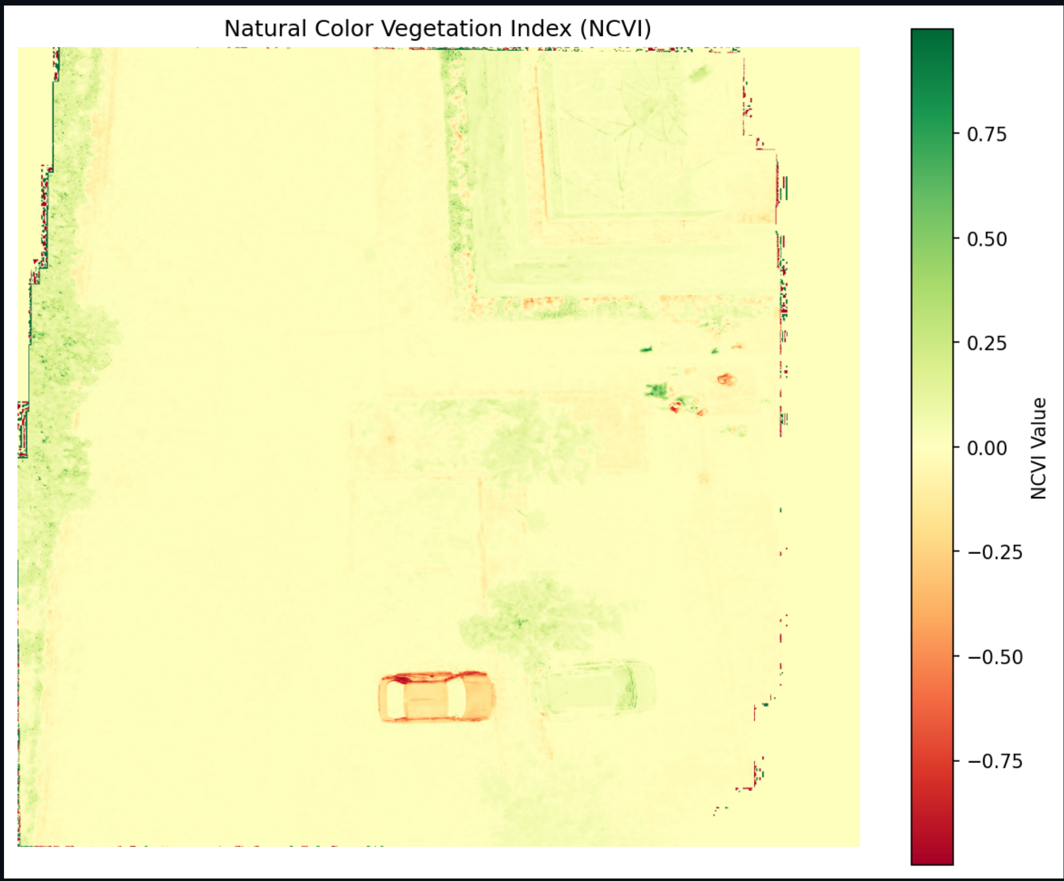


*Figure 4 — NCVI value distribution histogram and analytical plot.*

## 6. Results & Discussion

| Metric | Value |
| --- | --- |
| Images Processed | 52 (4000×3000 px) |
| Ground Sampling Distance | ≈ 2.4 cm/px |
| Orthomosaic Area | 0.42 ha |
| SfM Points (post-filter) | 55 534 |
| PLY File Size | 1.5 MB |
| NCVI Range | [−0.32, +0.41] |
| Coordinate System | EPSG:4326 (WGS 84) |

The pipeline successfully processed all 52 images in under 15 minutes on a consumer-grade laptop (Intel i7, 16 GB RAM). The SfM reconstruction produced a visually coherent point cloud that aligns well with the georeferenced orthomosaic when overlaid in the 3D viewer. The NCVI analysis correctly identified tree-lined streets, park areas, and urban impervious surfaces, demonstrating that meaningful vegetation assessments can be performed without multispectral sensors.

Limitations include the sparse nature of the SfM cloud (55K points vs. millions from commercial software like Pix4D), which is attributable to the use of a simple incremental pipeline without bundle adjustment. Future work should integrate scipy.optimize.least_squares for global bundle adjustment and dense multi-view stereo matching to achieve LiDAR-comparable point densities.

## 7. Conclusion

This report has presented a fully open-source, Python-based drone image processing pipeline that produces three complementary geospatial products: a georeferenced orthomosaic, a 3D SfM point cloud, and a vegetation index heatmap. The results are delivered through an interactive web portal (AeroVue) that enables real-time exploration of the data in both 2D and 3D. The methodology is reproducible, extensible, and suitable for educational and research applications in GIS, remote sensing, and precision agriculture. All source code, data, and the interactive portal are publicly available at https://norbertmuzila.github.io/drone-image-processing/.

*— End of Report —*